

Aadarsh Prajapati

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ABOUT

Full Stack Software Engineer specializing in building scalable and high-performance applications.

EDUCATION

- Amrita Vishwa Vidyapeetham** Kerala, India
• *Bachelor of Technology Computer Science and Engineering* Sept 2022 – Aug 2026
Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Distributed Systems, Deep Learning, Neuroengineering, Computer Security, Machine Learning, Networking, Databases
- AIFT Mumbai University** Mumbai, India
• *BA in Film, T.V and New Media Production* Aug 2021 – Sept 2022
Courses: Film Making, Editing and VFX

SKILLS SUMMARY

- **Languages:** Python, C++, C, SQL, JAVA, Javascript and Typescript
- **Frameworks:** Node.js, Express, Fastify, Next.js, React.js, Spring Boot,
- **Tools:** Docker, Redis, MongoDB, WebRTC, GIT, PostgreSQL, MySQL, SQLite
- **Area of Interest:** DevOps, System Design, Data Science, Rust, Web3
- **Skills:** FullStack application development and deployment, Leadership, Event Management, Filmmaking

EXPERIENCE

- **Software Development Engineer Intern - pracflow.ai** Remote
Fullstack Developer (Full-time) Nov 2024 - Feb 2025
 - **Digital Resource Platform:** Built a scalable platform for selling and sharing digital resources with secure payments.
 - **Optimized Storage:** Reduced redundant S3 uploads via resource-link mapping, cutting costs and improving UX.
 - **Version Control:** Enabled dynamic version updates without duplication, ensuring seamless access to the latest resources.
 - **Automated QR Passes:** Generated workshop passes with QR codes for streamlined operations.
 - **Team-Based Access Control:** Restricted client data edits to team members only, enhancing security and privacy.
 - **Resource Sharing:** Allowed therapists to price, manage, and securely share digital resources.
 - **Search & Filters:** Improved resource discovery with better search and filtering.
 - **MongoDB Optimization:** Resolved data inconsistencies and boosted performance with Aggregation Pipelines.
 - **Automated Sitemap Generation:** Dynamically generated therapist websites with indexed content for better SEO.
 - **Revenue Tracking:** Developed earnings dashboard for therapists selling resources.
- **Amfoss Amrita** On site
Full-Stack Engineer (full-time club member) Nov 2022 – Aug 2023
 - **Full-Stack Development:** Built high-performance, secure, and scalable applications with cost-efficient solutions.
 - **Automation Bots:** Developed custom bots to automate daily tasks, boosting productivity and reducing manual effort.
 - **Codebase Optimization:** Refactored code for performance, security, and maintainability while optimizing deployments.

PROJECTS

- **Olabs2.0 – National level hackathon winner:** A highly scalable and efficient learning platform integrating real-time collaboration, AI-driven assistance, and interactive coding tools. Built using open-source SFU servers for seamless video communication and WebSockets for low-latency interactions. [view project](#)
 - **Skill Development Section:** Integrated a feature-rich learning environment with:
 - * **Group Video Call (Shareable links, QR-based Access)** –Enabled seamless teamwork with group video calls, powered by open-source SFU servers for high scalability and low-latency communication, with QR-based and link based access.
 - * **Live Coding** – Real-time execution, debugging, and hands-on coding practice.
 - * **ML Visualizers** – Interactive tools for visualizing machine learning algorithms and data.
 - * **AI-Powered Assistance** – OLabs Mentor Chatbot for context-aware guidance.
 - * **Interactive Whiteboard** – 20+ tools and features for collaboration.
- **Exchange Platform – Scalable Microservices Architecture:** A high-performance exchange platform leveraging microservices, real-time data processing, and efficient traffic management. [github](#)
 - **Scalable Microservices Architecture:** Designed a modular system with independent services for handling traffic, data exchange, and processing.

- **Real-time Data Exchange:** Implemented WebSocket services to facilitate low-latency communication between users and the platform.
- **Dynamic Data Processing Engine:** Built an Engine service to process incoming data streams efficiently, ensuring seamless market operations.
- **Event-Driven Communication:** Integrated PubSub mechanisms to enhance inter-service communication and maintain system scalability.
- **Optimized Deployment:** Containerized services using Docker, enabling easy deployment, scaling, and maintenance.
- **Market Simulation:** Developed a Market Maker service to simulate market conditions for testing and optimization.
- **Object Detection – Deep Learning for Computer Vision:** Developed a deep learning-based object detection system leveraging an AlexNet-inspired CNN architecture for high-accuracy image classification and detection. **collab**
 - **Data Preparation and Augmentation:** Utilized a Kaggle dataset with diverse object categories. Applied augmentation techniques such as shear, zoom, and horizontal flip to improve model generalization.
 - **AlexNet-Inspired CNN Architecture:** Designed a five-layer convolutional network with varying filter sizes, batch normalization, and dropout layers to enhance feature extraction and model stability.
 - **Optimization and Hyperparameter Tuning:** Implemented Adam optimizer, early stopping, and learning rate scheduling to improve convergence and mitigate overfitting.
 - **Performance Evaluation:** Trained the model for 35 epochs, achieving a **training accuracy of 95.0%** and **validation accuracy of 44.8%**, with final training and validation losses of **0.1434** and **3.5979**, respectively.
 - **Challenges and Enhancements:** Addressed dataset imbalance, refined detection accuracy, and explored transfer learning with pre-trained models for improved performance.

ACHIEVEMENTS AND CERTIFICATIONS

- **Winner – OLABS Hackathon 2025, National level:** **devfolio**
- **First Prize and Global Nomination – NASA Space App Challenge 2024 (Kollam, India):** **project live**
- **Global Nomination – NASA Space App Challenge 2022 (Kathmandu, Nepal):** **project**
- **Participation – International Conference on Tsunami Risk Reduction and Resilience (ICTR3):** **details**
- **Full Stack + DevOps + System Design Cohort:** **certificate**

EXTRACURRICULAR ACTIVITIES

- **Team Leader – Short Film Production** Led a team of 30+ members in film production, managing scripting, direction, and execution.
- **Team Leader – Hackathon Winner (OLABS Virtual Lab)** Led a team in developing an AI-powered virtual lab platform, winning a national-level hackathon.
- **Team Leader – NASA Space App Challenge (First Prize and Global Nomination, Kollam)** Led a team of four developers, achieving global nomination for an innovative NASA data solution.
- **Organizer – Cybersecurity Awareness Program** Conducted a session for 9th-grade students on phishing risks and safe internet practices.